Prompt 1: Optimizing School Curriculum

A scheduling system for 626 students, 54 teachers and 59 rooms. Student walking falls by 81% and idle periods between lessons by 98%. All ten hard constraints are verified by an independent checker, and the timetable can be repaired locally when a room closes or a teacher is absent. The timetable is identical in every week.

−81% reduction in student walking per week

−98% reduction in idle periods between lessons

0 violations of the ten hard constraints (independent check)

10/10 injected faults detected by the checker

A Problem Statement and Scope

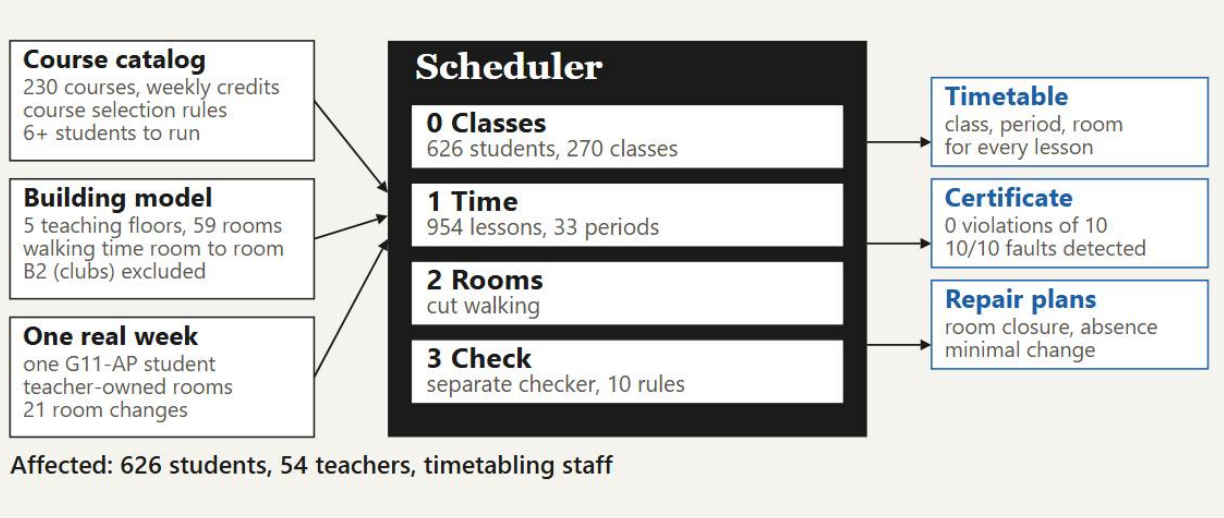
626 students, 54 teachers, 59 rooms

The brief requires an automatic school timetable, evidence that it improves on the current one, and verification that it satisfies every constraint. The inputs are the course catalog, a model of the building and one observed student week. A simulated school of 626 students (grades 9, 10 and 11 AP) is constructed from these inputs, scheduled automatically and compared with an emulation of the current method. **The timetable is identical in every week**; one week is planned and repeated.

BRIEF · Question.txt, verbatim

> The school has to produce a timetable that takes care of teachers, students, classrooms and courses all at once. This timetable affects the daily study and work of teachers and students, so having a high-quality timetable is important. We want you to consider the constraints in the school's timetable and optimize the timetable. By this, you should design a system and be able to explain why it is better than what we currently have. An example of a constraint is class hours: you may change Math class with English class, but the hours spent on math class in a semester should not change. The result should be a system that automatically schedules timetables. There is no standard answer. How the current system works, which constraints must be satisfied, which aspects are worth improving: all of these are for you to research, judge and trade off, and you should explain your reasoning in your submission. During the presentation, the judges will ask why you made these decisions.
>
> **2. Evaluating Your Results**
>
> This problem cares more about which aspects you think can be improved and how you improve them. Your submission should answer:
>
> - What problems in the current timetable system did you identify?
> - Which aspects did you choose to optimize (for example, minimizing student movement between classrooms)? How did you optimize them?
> - Who do these aspects affect, and how much? Why do they really matter?
> - Compared with the current timetable, how much does your solution improve things? How do you measure it?
> - How do you confirm that your timetable satisfies every constraint it must satisfy?
> - Can your system still cope when circumstances change? How to present your results most convincingly is also up to you.

Q1	Problems in the current timetable → **B**
Q2	Aspects optimized and method → **C, D**
Q3	Affected groups and magnitude of effect → **D**
Q4	Improvement and its measurement → **D**
Q5	Verification of the constraints → **E**
Q6	Response to changed circumstances → **F, G**



Affected: 626 students, 54 teachers, timetabling staff

Fig. 1 — *System context: inputs, scheduler stages and outputs.*

Data Sources and Simulation Scope

Real inputs: the course catalog and its selection rules, the building model with walking times between rooms, and one observed student week.

Simulated components: the 626 students and their course choices, the 54 teachers, and the emulated current method. All components are generated from a fixed random seed and are reproducible.

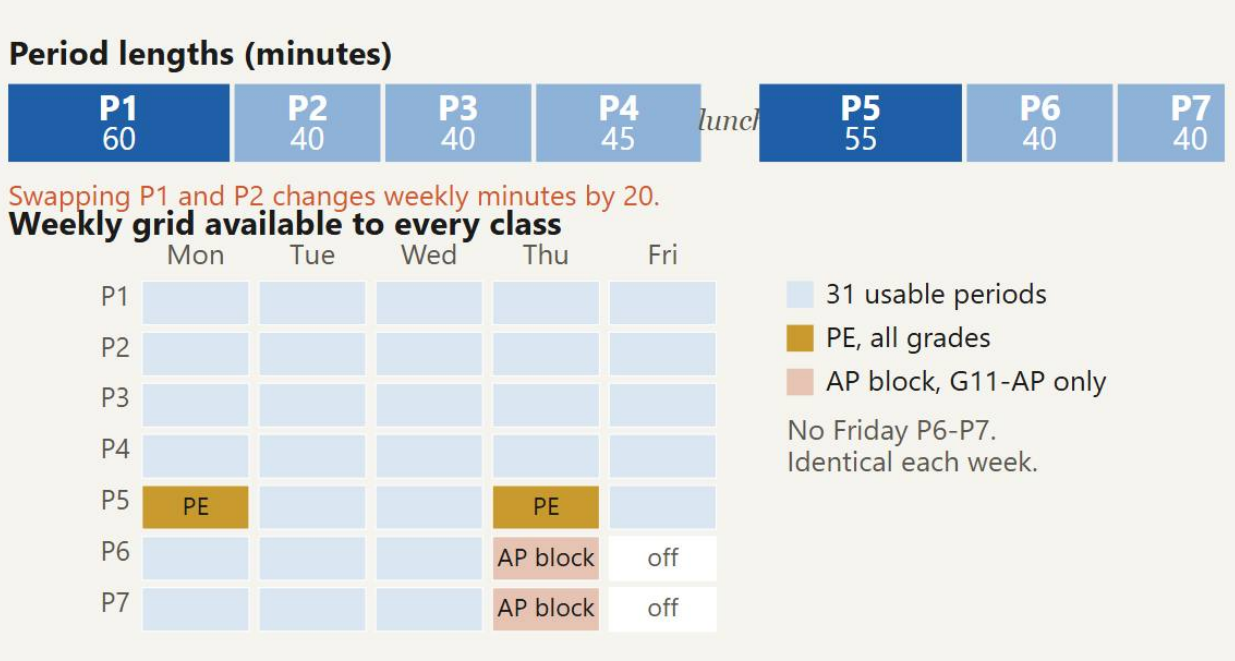


Fig. 2 — *The 33 periods available per week and their lengths.*

Contact Hours Measured in Minutes

Period 1 lasts 60 minutes and period 2 lasts 40. Moving a lesson between them preserves the lesson count but changes the weekly contact time by 20 minutes. Because the brief requires class hours to remain unchanged, contact time is counted in minutes.

Some periods are unavailable: PE occupies Monday and Thursday period 5, and Friday has no afternoon periods. The same grid applies in every week.

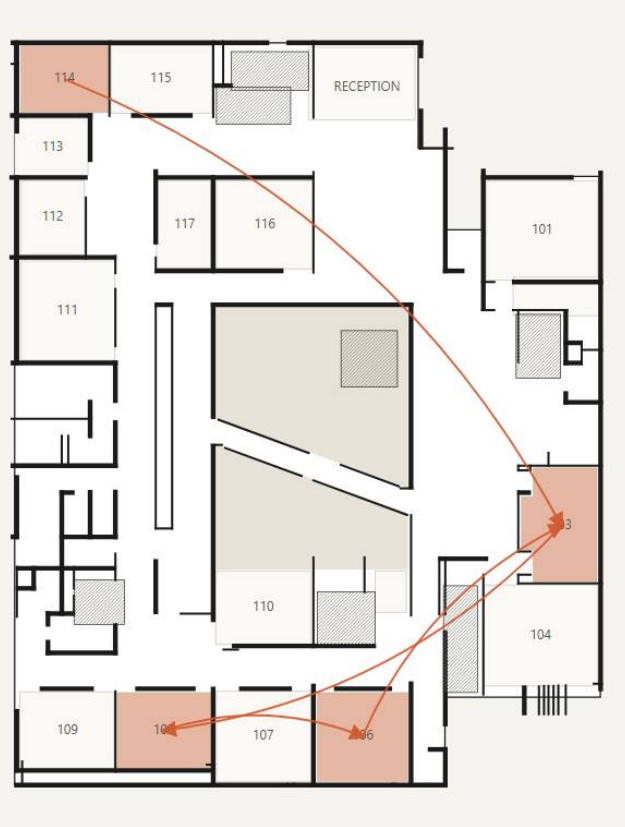
Assumption on Rooms

Floor B2 is used for club activities only and hosts no classes. It is removed from the room pool, which also removes all 14 laboratories: the sciences meet in ordinary classrooms and keep their practical double period. The pool is 49 classrooms (floors 1F to 4F) and 10 arts rooms (B1). The checker rejects any lesson placed on B2.

B Analysis of the Current Timetable

Question 1

In the observed week, a student changes room 21 times, because each teacher retains a room and students move between them. Free periods are scattered between lessons, and no check ensures that a lesson retains its weekly minutes when it is moved. Figures 3 to 5 document each problem.



Observed Monday: 114 → 103 → 108 → 108 → 106 → 106 → 103
4 room changes in one day; **21** per week, 488 s of walking.

Fig. 3 — *Observed Monday of one student on the ground-floor plan; each arrow is a room change. Each teacher owns a room, so the student moves between teachers' rooms throughout the day.*

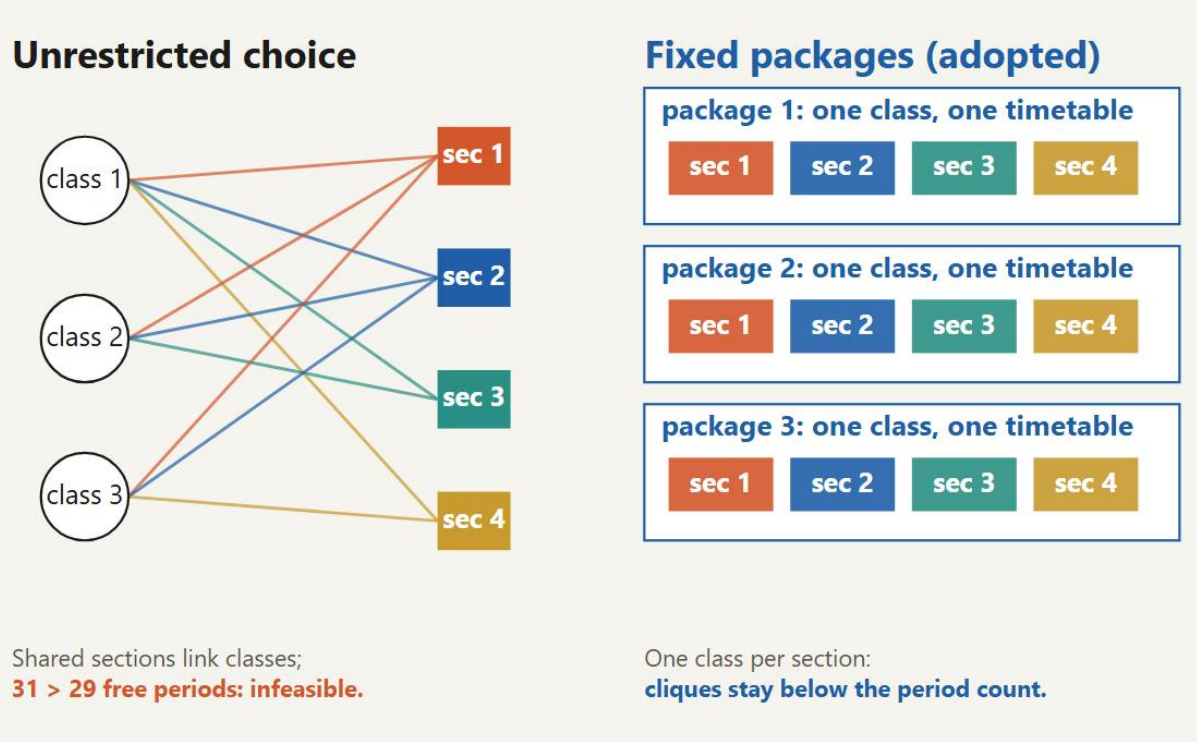

Unrestricted choice
class 1
class 2
class 3
sec 1
sec 2
sec 3
sec 4
Fixed packages (adopted)
package 1: one class, one timetable
package 2: one class, one timetable
package 3: one class, one timetable
sec 1
sec 2
sec 3
sec 4
Shared sections link classes;
31 > 29 free periods: infeasible.
One class per section:
cliques stay below the period count.

Fig. 4 — *Left: unrestricted elective choice. Right: students choose from fixed packages.*

Effect of Free Elective Choice on Feasibility

When students combine electives freely, each class shares students with many others, and all such classes require distinct time slots. In the simulated school this produces 31 mutually exclusive lessons for 29 free periods, so no valid timetable exists. A limited set of course packages is therefore offered; students in one package form a single class that takes all of its lessons together.

Weekly Contact Minutes per Class

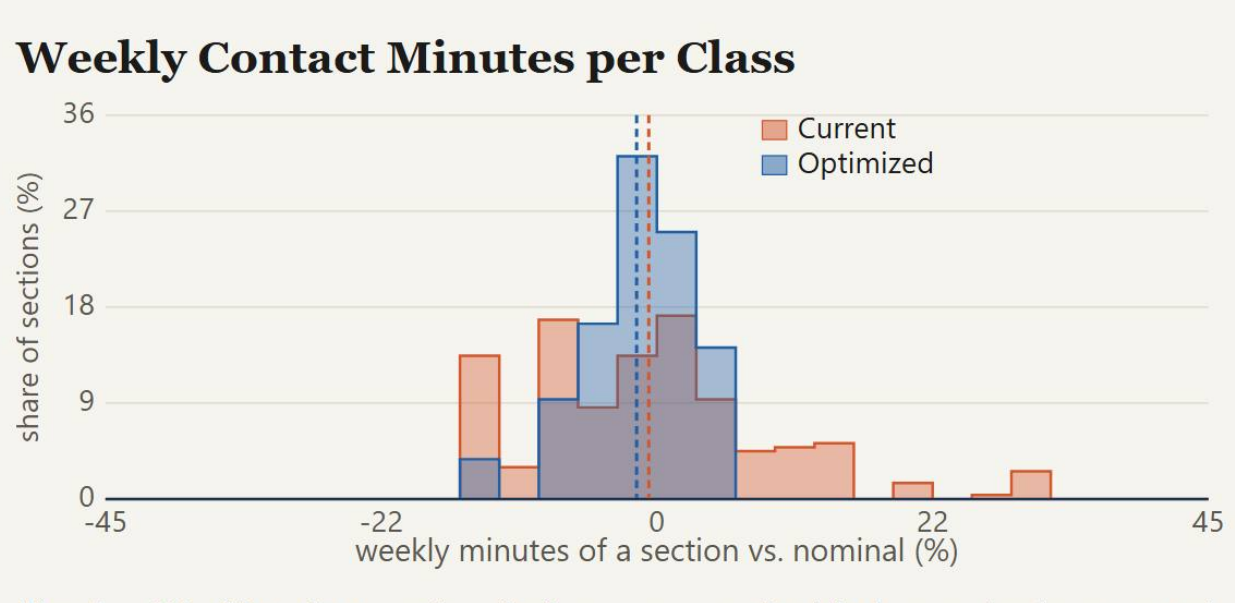


Fig. 5 — *Weekly minutes of each class compared with the nominal amount. A timetable constructed without regard to period length places* **110 of 270** *classes outside the ±8% band; the optimized timetable places none (the brief: class hours should not change).*

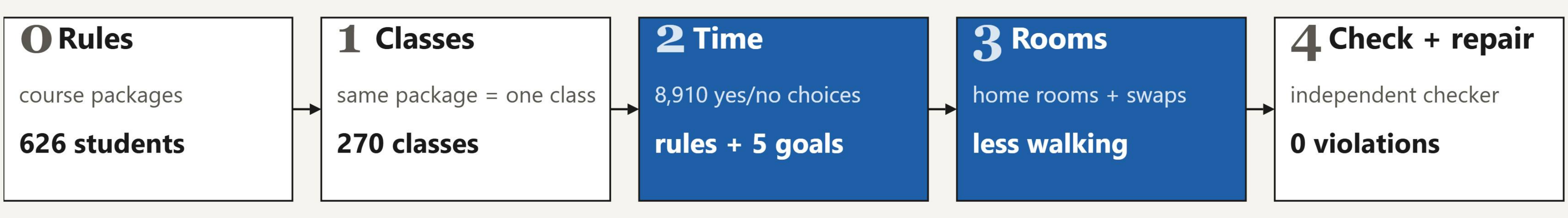
- No component of the current method targets student walking, idle periods between lessons or overloaded days.
- Repair after the loss of a room is performed manually.
- There is no independent verification that the timetable satisfies the rules.

Summary: high student walking, scattered idle periods and no check on contact hours.

C Method

Question 2

The method is divided into stages that can each be verified separately: grouping students into classes, assigning each class to periods, assigning rooms, and verifying all constraints on the finished timetable, with a local repair procedure. The equations below state the constraints and objectives used.



0 Rules
course packages
626 students

1 Classes
same package = one class
270 classes

2 Time
8,910 yes/no choices
rules + 5 goals

3 Rooms
home rooms + swaps
less walking

4 Check + repair
independent checker
0 violations

Fig. 6 — *The five stages and the size of the problem at each.*

0 Rules
Course credits are converted into weekly lessons. Selection rules determine the permitted course packages.

1 Classes
Students with the same package form one class of at most 24. A course runs only with at least 6 students.

2 Time
A constraint solver assigns a period to every lesson, subject to all hard constraints, while minimizing idle gaps. One binary variable per class and period keeps the model compact.

3 Rooms
Each class receives a home room. Remaining lessons are matched to free rooms and swapped to reduce walking.

4 Check
An independent program checks the ten constraints. After a change, only the affected lessons are re-planned.

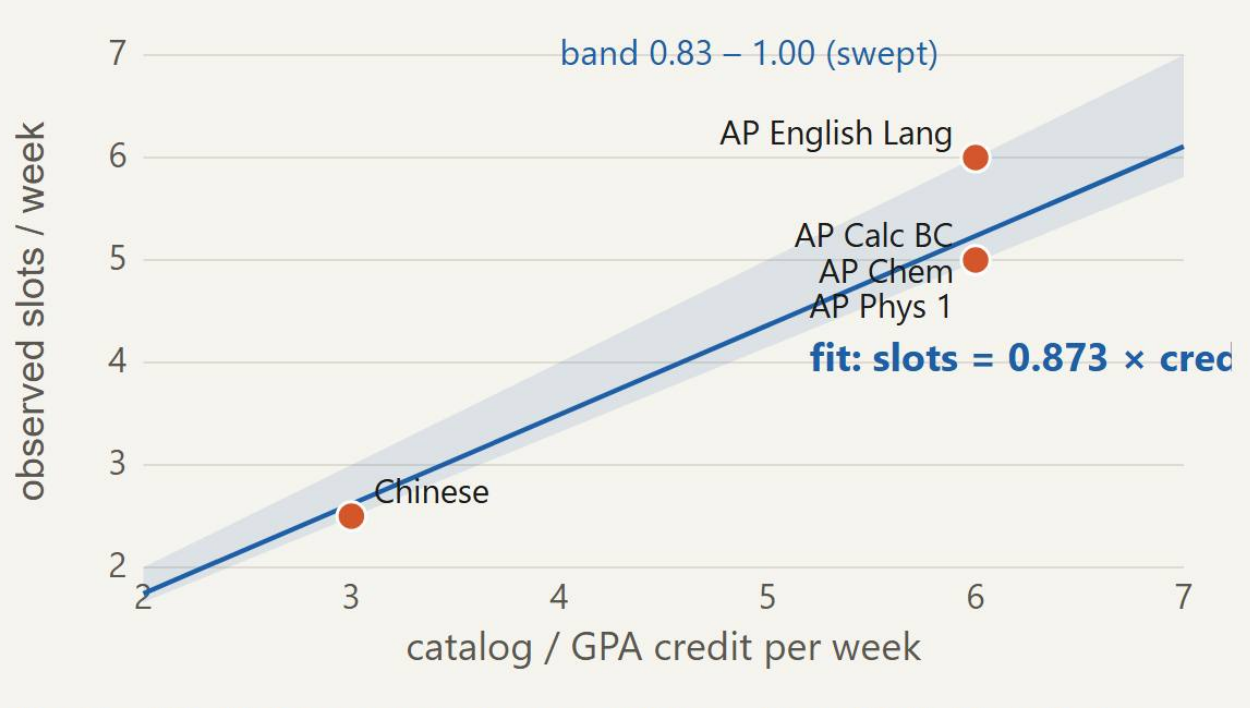


Fig. 7 — *Weekly lessons per course: catalog credit multiplied by a factor fitted on the observed week, evaluated over a band of values.*

CREDIT TO WEEKLY SLOTS

$$n_c = \left\lfloor k\kappa_c + \tfrac{1}{2} \right\rfloor, \quad k = \frac{\sum_i \kappa_i o_i}{\sum_i \kappa_i^2} \approx 0.873$$

HARD CONSTRAINTS ($z_{g,s} \in \{0,1\}$)

$$\sum_s z_{g,s} = n_g, \quad \sum_{g\in\sigma} z_{g,s} \le 1, \quad \sum_{g\in T_t} z_{g,s} \le 1$$

$$\sum_{g:\, a_g \ge a} z_{g,s} \le R_\kappa(a) \quad \text{(room area)}$$

CONTACT HOURS IN MINUTES

$$\left|\sum_s \mu_s z_{g,s} - n_g\bar{\mu}\right| \le \max(\tau n_g\bar{\mu},\, 9),\ \tau = 0.08$$

STAGE 1: SLOT OBJECTIVE

$$\begin{aligned}\min\ & w_{gap}\sum_\sigma c_\sigma\Gamma_\sigma + w_h\sum_\sigma c_\sigma H_\sigma + w_{sp}\sum_g |g|\,\mathrm{rep}_g \\ & + w_m \sum_{g\ \mathrm{hard}} |g| \sum_s (p_s - 1) z_{g,s} + w_{tg}\sum_t \Gamma_t\end{aligned}$$

STAGE 2: PERSON-SECONDS

$$\min_r \sum_{(a,b)\in\mathcal{L}_{stud}} n_{ab}\,\tau(r_a, r_b) + \lambda \sum_{(a,b)\in\mathcal{L}_{teach}} \tau(r_a, r_b)$$

SCORE, FAIRNESS, ROBUSTNESS

$$J = \bar{\ell}_{stud} + \alpha\bar{L}_t + \beta\,\mathrm{CVaR}_q(L), \qquad \mathbb{E}[\Delta] = \sum_\xi p_\xi\,\Delta(\xi)$$

Students remain in their rooms and teachers move: λ is the weight of one second of teacher walking relative to one second of student walking. Γ = idle periods between lessons, H = days with six or more classes, rep = a class meeting twice on one day.

Summary of the constraints and objectives: each class receives exactly its number of lessons; no student, teacher or room is assigned to two places at once; weekly minutes remain near the nominal amount; idle gaps and walking are then minimized.

D Results and Comparison with the Current Method

Questions 3 and 4

Gains are measured against 24 timetables generated as the school generates them today (any valid timetable, teacher-owned rooms), on the same school. Lower values are better in every chart. Floor plans show where students change rooms, histograms show the effect per student, and teacher grids show where idle periods fall.

🟥 Current method (24 emulated timetables calibrated to the observed week; plots show run 0) 🟦 Optimized

Student Movement by Floor

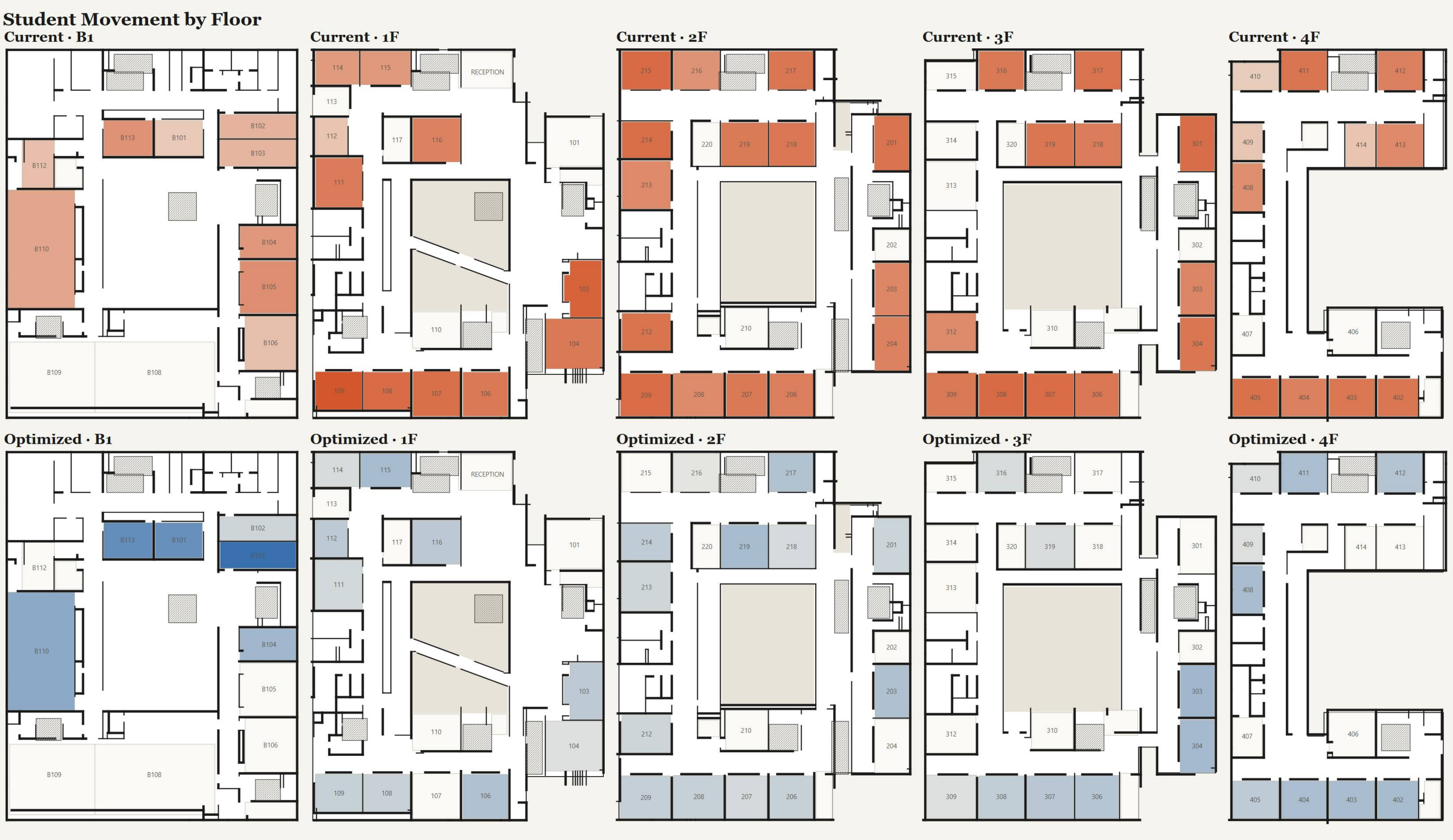


Fig. 8 — *Building model, five teaching floors, to scale (B2, reserved for clubs, is excluded). Darker rooms are touched by more room changes per week; the scale is common to both rows. With teacher-owned rooms, movement is distributed across the building; with home-class rooms it remains near each class.*

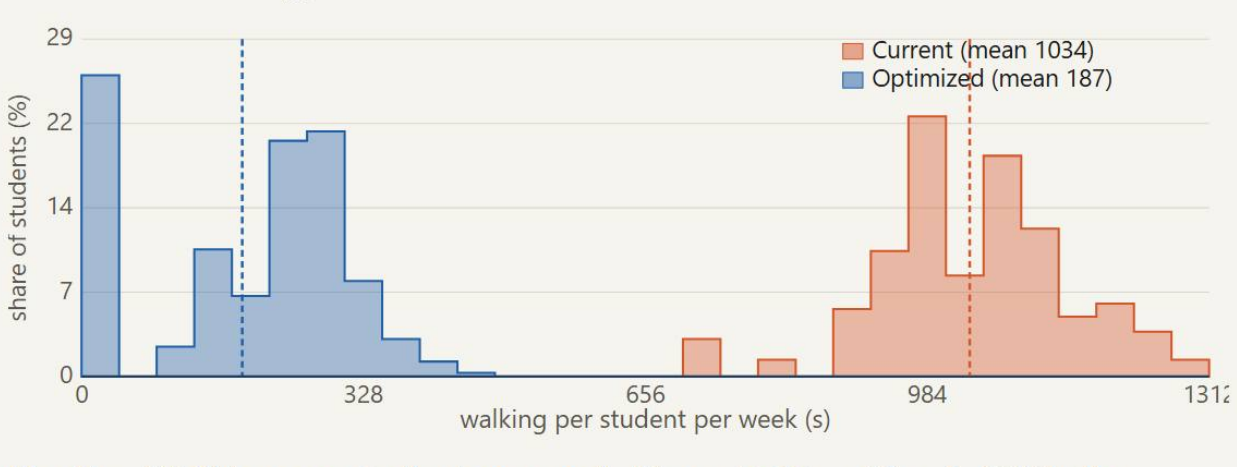


Fig. 9 — *Walking per student per week. Mean 1006 → 187 s (−81%); the worst-20% tail falls too, so the gain is not concentrated in a few students.*

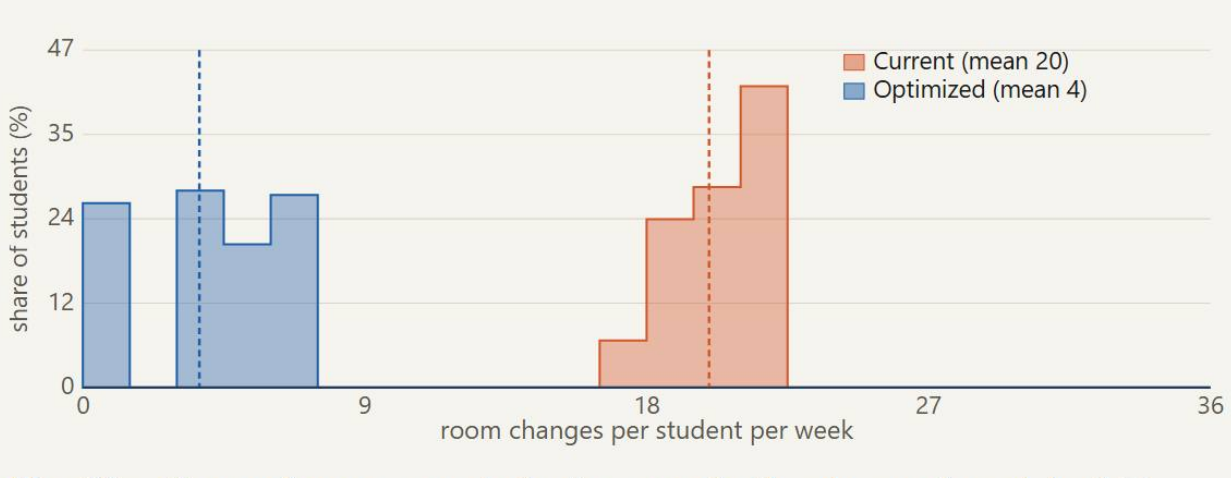


Fig. 10 — *Room changes per student per week; the observed week had 21.*

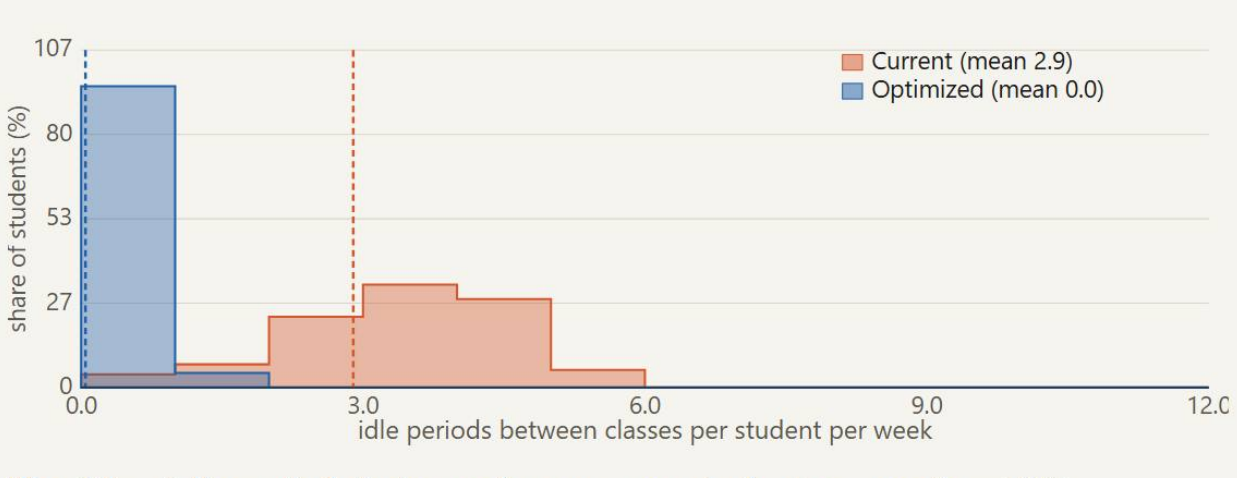


Fig. 11 — *Idle periods between lessons, per student per week: −98%.*

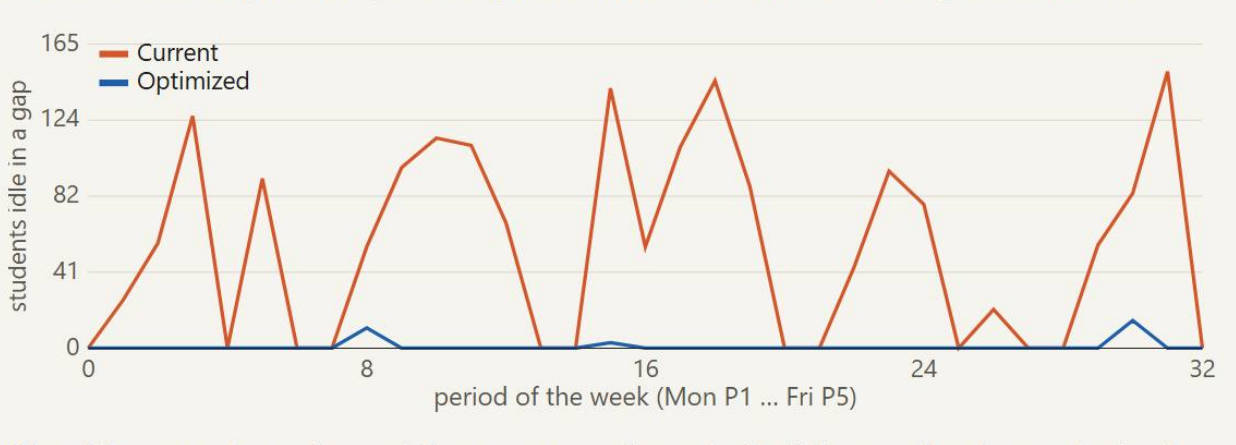


Fig. 12 — *Students in an idle gap at each period of the week. The optimized timetable moves idle periods to the start or end of the day.*

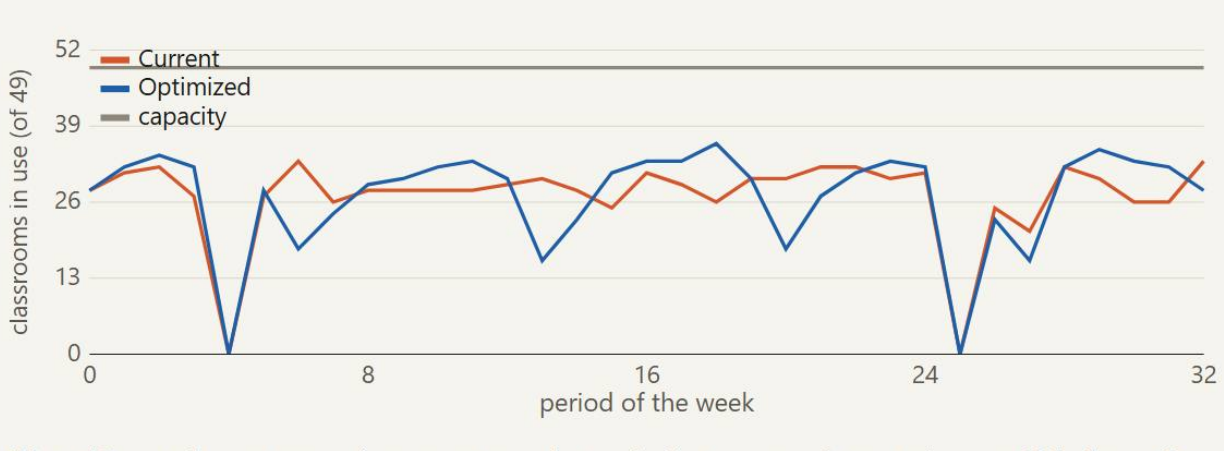


Fig. 13 — *Classrooms in use at each period. Demand remains well below the 49 available, so the saving does not result from using fewer rooms.*

Contribution of Room and Time Decisions

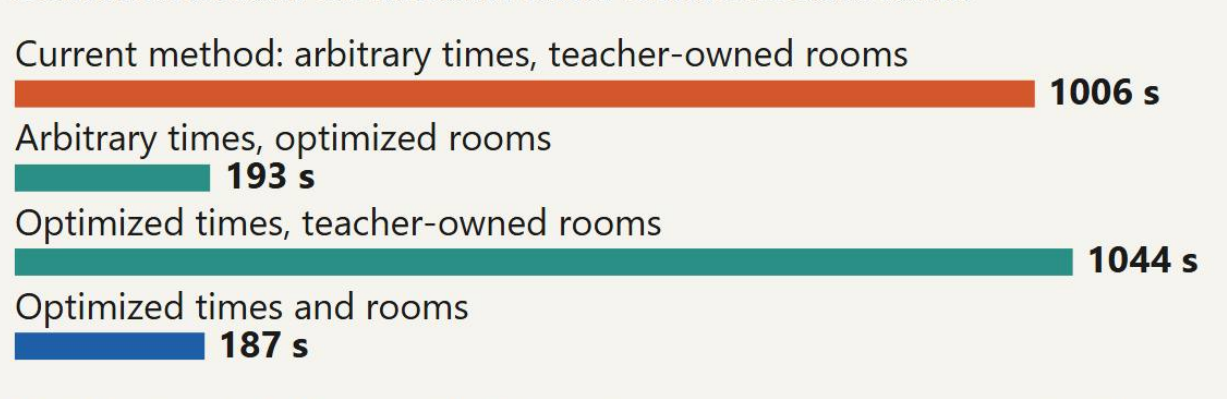


Fig. 14 — *Each decision applied alone. Room assignment accounts for the reduction in walking; time assignment accounts for the reduction in idle gaps.*

Effect on Each Group

WHO	MEASURE (PER WEEK)	CURRENT	OPTIMIZED
626 students	walking (s)	1006	187
	room changes	20.1	3.7
	idle periods	2.78	0.05
	worst-20% walking (s)	1158	331
	hard subjects: mean period	3.75	3.31
54 teachers	idle gaps	6.6	2.9
	walking (s)	442	509
	schedule loss L	12.1	11.6
Everyone	classes outside the minute band	110	0

Notes on the Results Table

- Orange denotes the current method and blue the optimized timetable, for the same school and lessons.
- Students improve on every measure.
- Teachers walk 15% more, by design: it is more efficient for 54 teachers to move than for 626 students.
- Teacher idle periods change by -56%.

Teacher Schedules

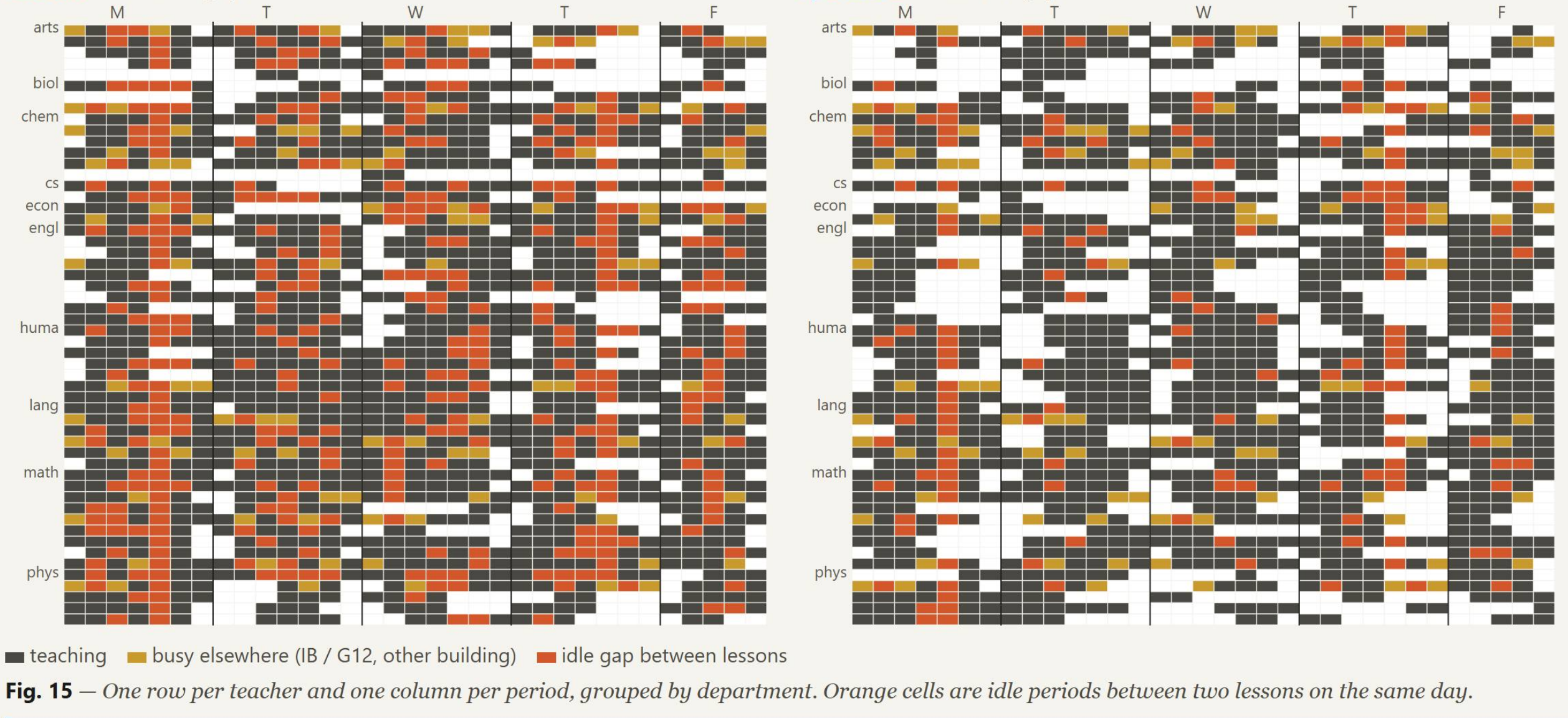


Fig. 15 — *One row per teacher and one column per period, grouped by department. Orange cells are idle periods between two lessons on the same day.*

Summary: student walking falls by 81% and idle periods by 98%; teacher walking changes by +15%.

E Constraint Verification

Question 5

An independent program, sharing no code with the scheduler, reads the finished timetable and checks ten constraints. To confirm that the checker detects violations, ten faults are deliberately introduced, one per constraint, and each must be reported.

RULE	HARD CONSTRAINT	OPTIMIZED	CURRENT
H1	section meets exactly its weekly slot count, on distinct slots	0	0
H2	slot exists, is not a fixed block for the grade, not teacher-external-busy	0	0
H3	no student in two places at once	0	0
H4	no teacher double-booked; teacher qualified for the department	0	0
H5	room exists (not on excluded floor B2), right kind, big enough, not double-booked	0	0
H6	teacher consecutive-period cap	0	0
H7	lab sections have a double period	0	0
H8	weekly minutes within tolerance of nominal (contact-hours invariant)	0	110
H9	each student receives every course's stated weekly slots	0	0
H10	every consecutive-period room change reachable inside the break	0	0

Fig. 16 — *270 sections, 626 students, 54 teachers, 720 room legs checked. Current = average over 24 simulated timetables.*

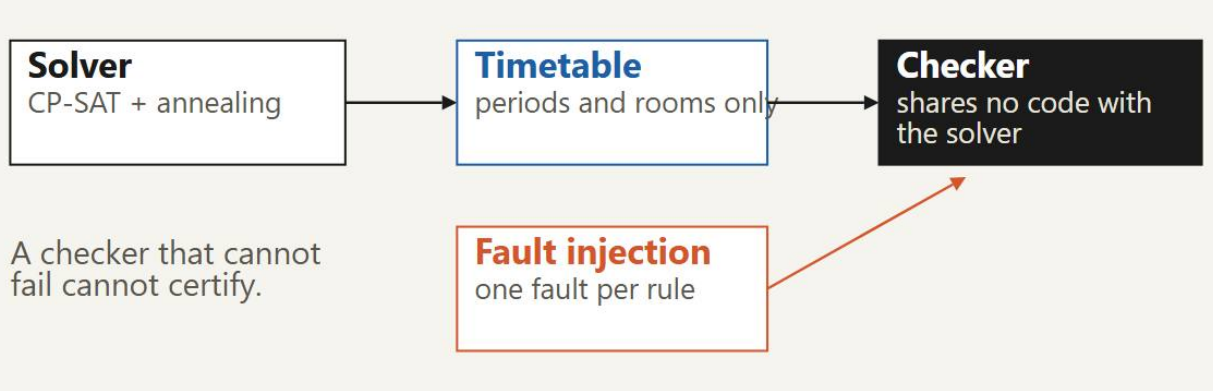


Fig. 17 — *The checker operates only on the finished timetable. The fault-injection test shows that it can fail: all 10 injected faults were detected.*

The Ten Hard Constraints

- Every class receives all its weekly lessons, in distinct periods.
- Lessons avoid PE time and the teacher's external commitments.
- No student, teacher or room is assigned twice in one period.
- The room has the correct type and sufficient area; the teacher is qualified for the department.
- Teachers do not exceed their consecutive-period limit; laboratory classes receive a double period.
- Weekly minutes remain within tolerance of the nominal amount.
- Every room change is reachable within the break between lessons.

20 automated tests additionally cover loads, course rules, the solver, repairs and the observed week.

F Sensitivity Analysis

Question 6

Several inputs of the simulation are assumptions. Each was varied individually and then several together, and the full pipeline was re-run. The walking reduction remains within a narrow band in nearly all cases. The exceptions are genuine limits: a much heavier timetable, or the loss of many rooms, can leave no valid timetable.

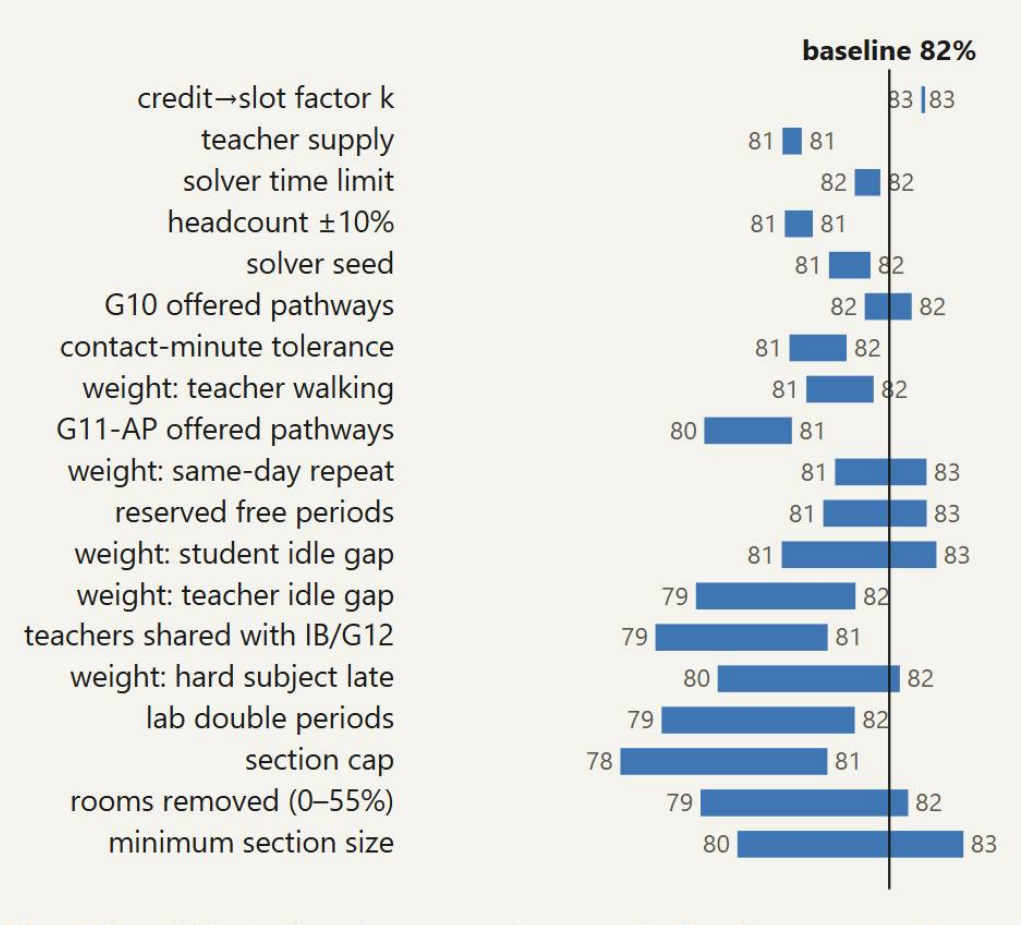


Fig. 18 — *Effect of each assumption, varied on its own, on the walking reduction (%); 62 runs, each solved and checked again (57 with no violated constraint).*

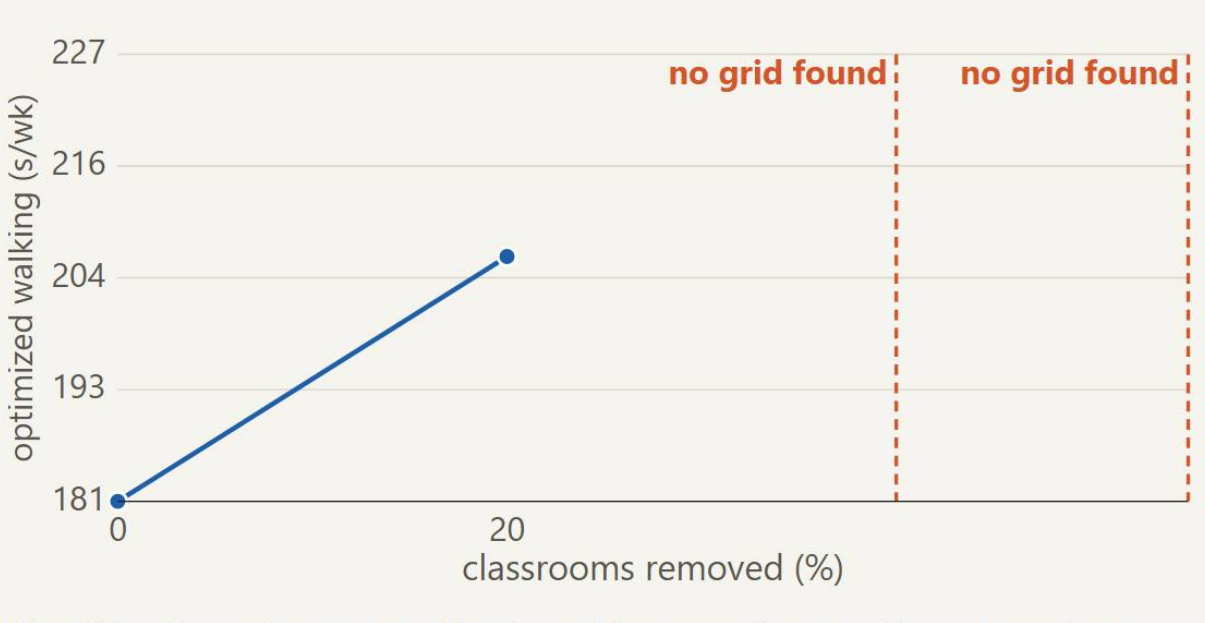


Fig. 19 — *Loss of rooms: valid timetables were found with up to 20% of rooms removed and none with 40% or more; the program reports infeasibility instead of an invalid timetable.*

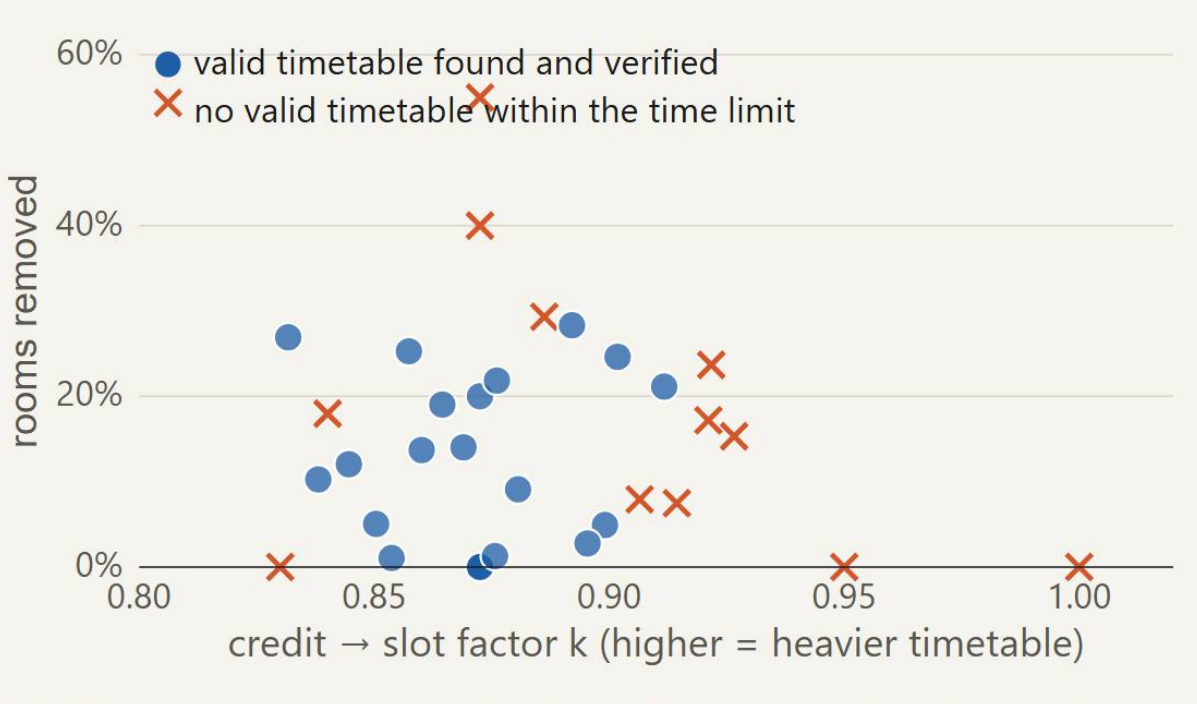


Fig. 20 — *Operating envelope. Each marker is one run; the timetable load (k) and the share of rooms lost determine whether a valid timetable is found.*

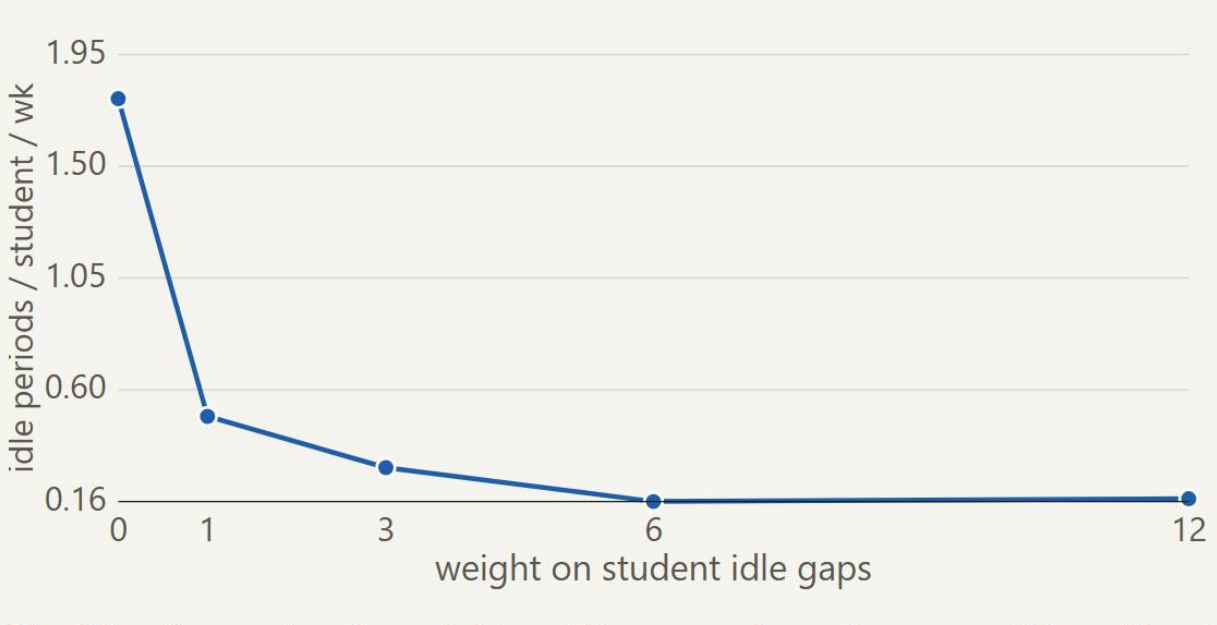


Fig. 21 — *Increasing the weight on idle gaps reduces them smoothly, without a discontinuity.*

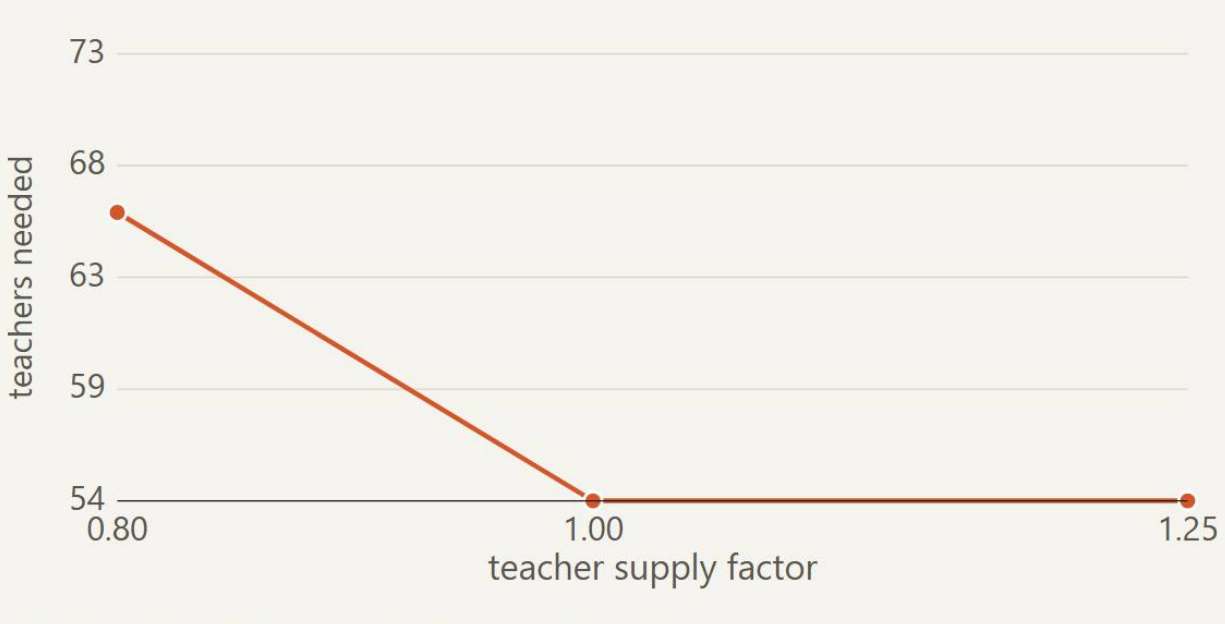


Fig. 22 — *Teachers required as supply tightens.*

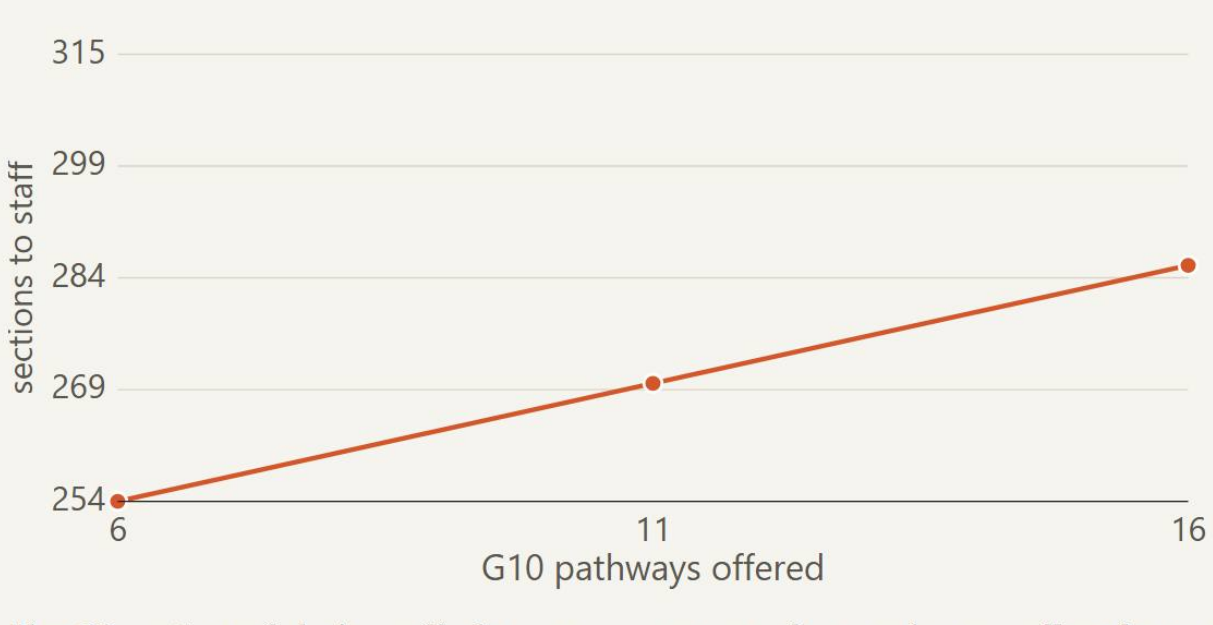


Fig. 23 — *Cost of choice: offering more course packages gives smaller classes and more classes to staff.*

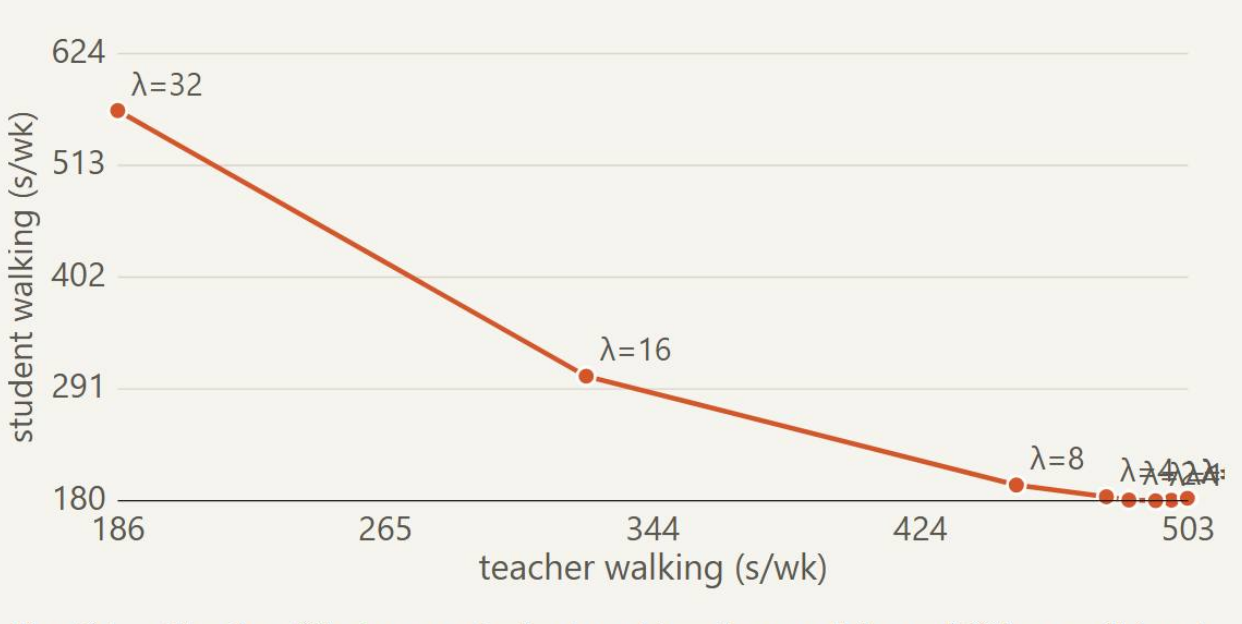


Fig. 24 — *Trade-off between student and teacher walking: shifting walking to teachers is inexpensive until teacher walking is weighted heavily.*

Summary of Sensitivity Findings

- The walking reduction remains between **78%** and **83%** across all single-assumption changes.
- It is robust to the objective weights, class-size limits, staffing and the solver random seed.
- It is **not** robust to timetable load: a heavier load (k close to 1) or the loss of 40% of rooms leaves no valid timetable, for the current method as well as the optimized one.
- Offering more course packages increases the number of classes to staff.

G Robustness and Repair

Question 6

When a room closes, a teacher is absent or enrollment grows, the system applies the smallest possible change: it first seeks a substitute teacher, then moves the lesson to another room, and changes a lesson's time only as a last resort. Every repaired timetable is checked again.

Room 114 closed for the week (1F)

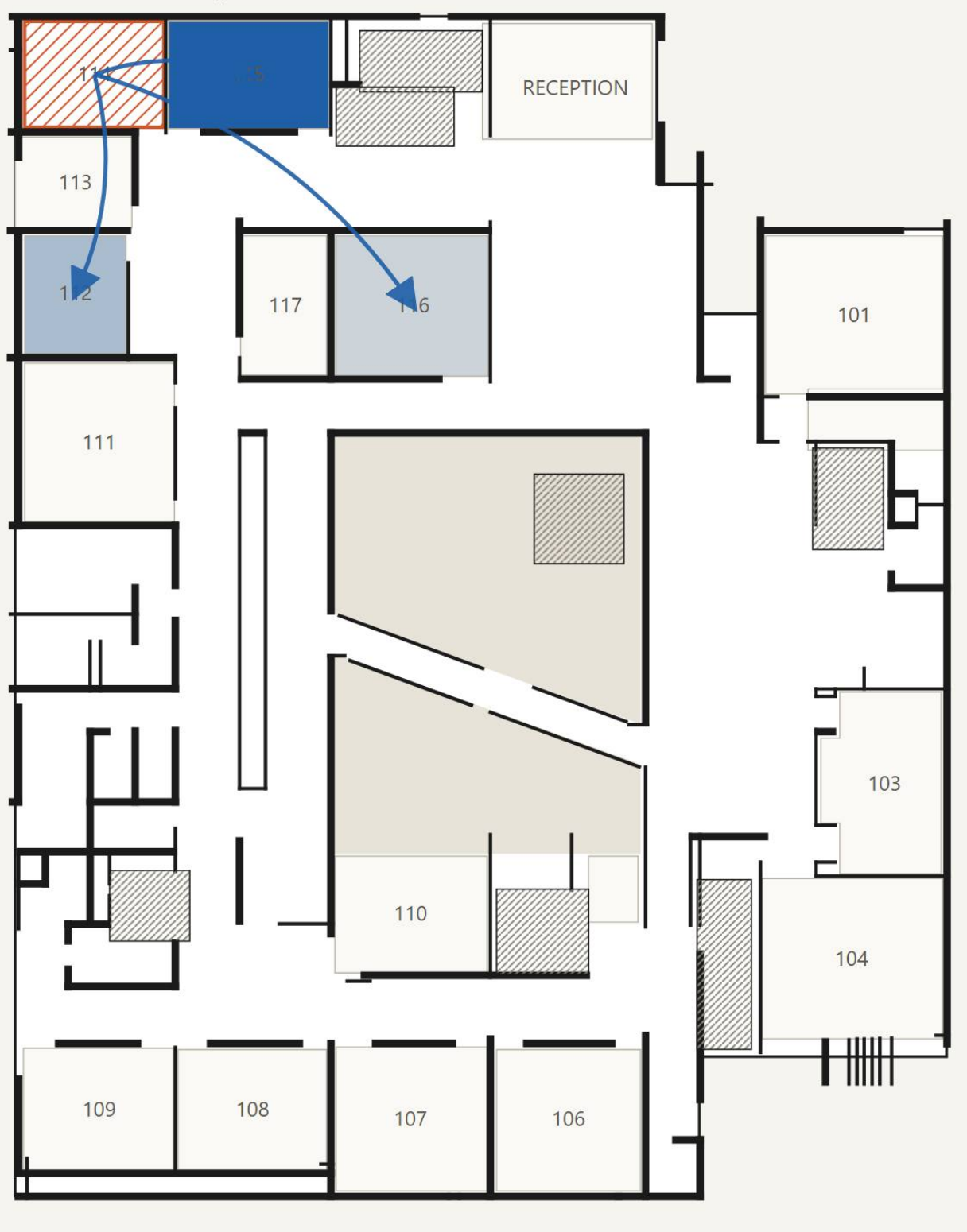


Fig. 25 — *27 lessons lose their room; 27 are re-roomed in a free room of the same kind,* **0 lessons change time**. *Blue = receiving rooms.*

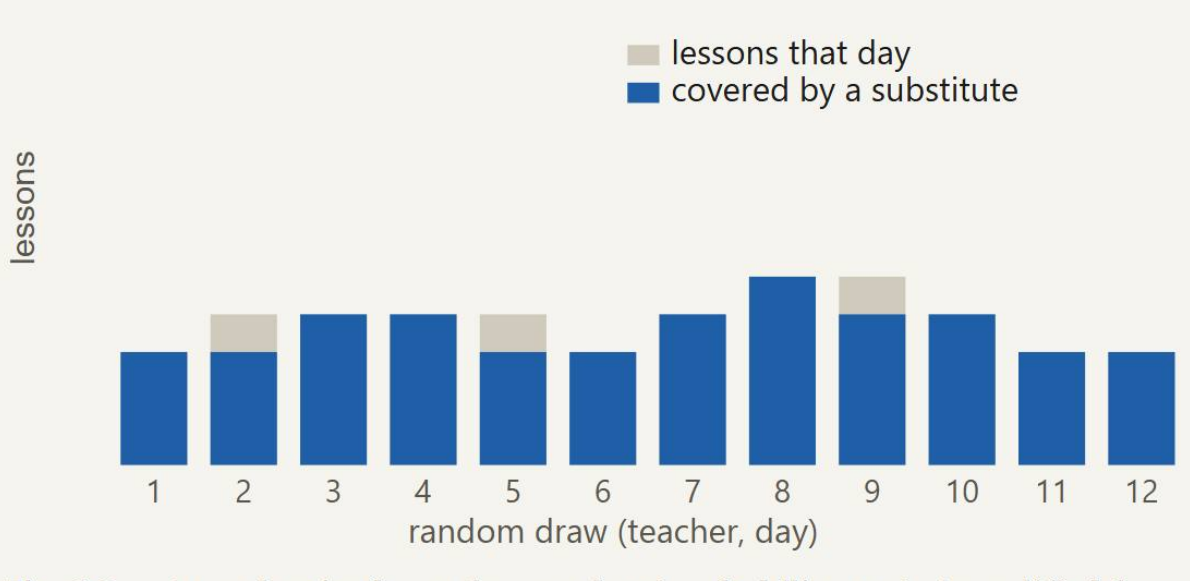


Fig. 26 — *A teacher is absent for one day (probability 0.4). A qualified free colleague covers 93% of lessons; no lesson moves.*

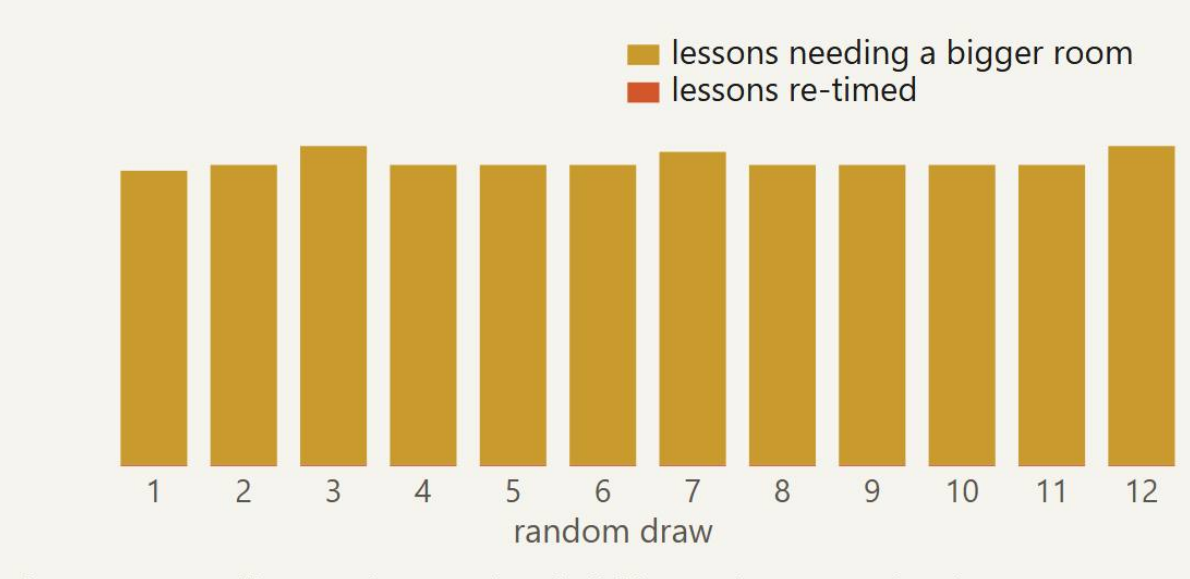


Fig. 27 — *Enrollment rises 5% (probability 0.3). Rooms that become too small are replaced by larger ones; lesson times are unchanged.*

Repair Procedure

- **1. Substitution.** A qualified teacher who is free in that period takes the lesson.
- **2. Room change.** The lesson keeps its period and moves to a free room of the correct type and size.
- **3. Time change.** Only if no room is free, the affected classes are re-planned while all other lessons stay fixed.

Over 12 random draws of each event, every repair was possible and no lesson required a change of time. Expected cost per event: **0.00** lessons moved and **20.9** room changes.

Summary: a closed room, an absent teacher or higher enrollment is handled by local changes, without rebuilding the timetable.

H Limitations and Reproducibility

scope of the evidence

- **Room assumption.** Floor B2 (club use only, including all laboratories) is excluded; sciences meet in classrooms.
- **One real week only.** The current system is an emulator calibrated to it (about 20 vs 21 changes); everything else is simulated. Registrar data can replace the simulated inputs.
- **Operating envelope.** Valid timetables were found up to k = 0.87 and with up to 20% of rooms removed; beyond k = 0.95 or 40% of rooms removed none was found in the time limit (Fig. 20).
- **Not proven optimal.** CP-SAT stopped at its time limit, 78% from its bound: "optimized" means best feasible found.
- **Pathway classes trade choice for feasibility** (Fig. 4); the price is quantified in F.
- **Minutes:** sections of one course can still differ by up to 36% in weekly minutes; the next rule to add is equality across sections.
- **Reproduce:** `python -m tt.run`, `python -m tt.sensitivity`, `python -m tt.poster2`.
- Every number on this poster is read from `output/*.json`; seeds and the config hash are fixed.

Constraints from the course catalog · building model: five teaching floors, 59 rooms (floor B2 excluded), stair-aware walking times · simulated school, seed 20261024

Python, OR-Tools CP-SAT